The precision penalty dominates unless the latency is sufficiently large. The upper constant is not claimed sharp, but even a rigorous intrinsic lower bound for this dyadic probe is

$$\kappa_{\mathrm{op}} \geq \frac{16\sqrt{2}}{7} \approx 3.23249. \tag{11}$$

Indeed $H_0 = I + A_F$ has operator norm 4 and $H_0 X = \sigma H_0 e_v h^T$ has Frobenius norm $2\sqrt{2}\sigma$. Thus, even replacing C_3 by this lower number does not remove the inverse-square penalty under the stated equal-per-action, aggregate-Frobenius guarantee policy. This is not a lower bound for every three-probe design, arbitrary noise allocation, or a physical measurement protocol.

The resulting scientific decision is to retain this construction as a query-count and norm-separation example, and prioritize the broader representation below. Further constant tuning on the triangle family is not the strongest next attack.

6 Arbitrary local frames and a scalar constraint Laplacian

Let G now be any simple q-regular graph. No coloring is specified. For each i, let $N[i]$ be its closed neighborhood, and let $V_i \in \mathbb{R}^{(q+1)\times q}$ contain the probe rows x_j^T, $j \in N[i]$. Assume V_i has rank q. Let ℓ_i be a unit generator of $\ker V_i^T$, indexed by $N[i]$. Its sign is immaterial. Assume all nonzero squared singular values of all V_i lie in $[\alpha, \beta]$ with $\alpha > 0$.

On $\mathcal{S}_G$ with Frobenius inner product define

$$(RH)_i = \sum_{j \in N[i]} \ell_{ij} H_{ij}.$$

Then $R^* a$ has diagonal $\ell_{ii} a_i$ and edge entries $(\ell_{ij} a_i + \ell_{ji} a_j)/2$. Put

$$Q = I - R^* R, \qquad K = I - R R^*.$$

Lemma 6.1 (Exact local-to-global identity)**.** *The symmetric matrix K has entries*

$$K_{ii} = \tfrac{1}{2} \sum_{j \sim i} \ell_{ij}^2, \qquad K_{ij} = -\tfrac{1}{2} \ell_{ij} \ell_{ji} \quad (ij \in E),$$

and

$$a^T K a = \frac{1}{2} \sum_{ij \in E} (\ell_{ij} a_i - \ell_{ji} a_j)^2. \tag{12}$$

For the measurement normal operator $N = M_X^* M_X$,

$$\alpha Q \preceq N \preceq \beta Q. \tag{13}$$

The measurement is injective on $\mathcal{S}_G$ if and only if K is positive definite.

Proof. Directly evaluating RR^* gives diagonal $\ell_{ii}^2 + \frac{1}{2}\sum_{j\sim i} \ell_{ij}^2 = (1 + \ell_{ii}^2)/2$ and edge entries $\ell_{ij}\ell_{ji}/2$. This proves (12). For the supported row h_i, the local frame bound is

$$\alpha(\|h_i\|_2^2 - |\ell_i^T h_i|^2) \leq \|V_i^T h_i\|_2^2 \leq \beta(\|h_i\|_2^2 - |\ell_i^T h_i|^2).$$

Sum over rows to obtain (13). The nonzero singular values of R give the same nontrivial eigenvalues of R^*R and RR^*, so $Q > 0$ iff $K > 0$. Alternatively, a vector with $Ka = 0$ gives a nonzero $R^* a$ in $\ker Q$, and the converse follows by applying R to a vector in $\ker Q$. Equation (13) completes the proof. □

Equation (12) is a rank-one cellular-sheaf Laplacian, with edge restriction maps $\ell_{ij}/\sqrt{2}$ and $\ell_{ji}/\sqrt{2}$. This class of Laplacians and its resistance theory are established [3]. In general it is not an ordinary unsigned graph Laplacian, nor necessarily an orthogonal-connection Laplacian. Algorithms for those narrower classes cannot be imported without checking their hypotheses.

6.1 An exact decoder with one scalar solve

Let J embed a symmetric supported matrix into the direct sum of its supported row spaces. This embedding is isometric for the Frobenius norm. Let $\Pi_i = I - \ell_i \ell_i^T$ and $\Pi = \mathrm{diag}(\Pi_i)$. Define a local data lift

$$(TY)_i = V_i (V_i^T V_i)^{-1} Y_i^T.$$

Then $T(M_X H) = \Pi J H$, $\|T\| \le \alpha^{-1/2}$, and $J^* \Pi J = Q$. The adjoint J^* averages the two copies of each off-diagonal entry. For $Z = J^* T Y$, solve

$$Ka = RZ, \qquad \widehat{H} = Z + R^* a. \tag{14}$$

The identity

$$Q^{-1} = I + R^* K^{-1} R \tag{15}$$

follows by multiplication. Thus (14) is exact on noiseless data. Local matrices have fixed size when q is fixed. The global solve has n scalar unknowns and graph sparsity, but its numerical complexity depends on the actual constraint matrix.

Theorem 6.2 (General Green-diagonal criterion)**.** *Assume $K > 0$, and put $g = \max_i (K^{-1})_{ii}$. Define*

$$\kappa_{\mathrm{op}} = \sup_{0 \neq H \in \mathcal{S}_G} \frac{\|H\|_{\mathrm{op}}}{\|HX\|_{\mathrm{F}}}.$$

Then

$$\boxed{\frac{\max\{1, g-1\}}{\beta} \le \kappa_{\mathrm{op}}^2 \le \frac{1 + (q+1)g}{\alpha}.} \tag{16}$$

The decoder (14) has arbitrary-noise amplification at most the square root of the upper bound. For fixed q and uniform positive local frame bounds, a family has uniformly bounded optimal operator amplification if and only if its g values are uniformly bounded.

Proof. For a unit vector u, put $A_u = \mathcal{P}_G(uu^T)$, so $u^T H u = \langle A_u, H \rangle$ for supported H. Hilbert-space duality gives

$$\kappa_{\mathrm{op}}^2 = \sup_{\|u\|_2 = 1} \langle A_u, N^{-1} A_u \rangle.$$

Inverting (13) and using (15) bounds this by

$$\frac{1}{\alpha} \sup_u \left[\|A_u\|_{\mathrm{F}}^2 + (RA_u)^T K^{-1} (RA_u) \right].$$

The first term is at most one. Also

$$\begin{aligned} \|RA_u\|_1 &\le \sum_i |u_i| \left(\sum_{j \in N[i]} u_j^2 \right)^{1/2} \\ &\le \left(\sum_i \sum_{j \in N[i]} u_j^2 \right)^{1/2} = \sqrt{q+1}. \end{aligned}$$

Positive definiteness implies $|(K^{-1})_{ij}| \le g$, which proves the upper bound.

For the lower bound, each functional $R_i(H) = e_i^T H \ell_i$ has absolute value at most $\|H\|_{\mathrm{op}}$. Therefore

$$\kappa_{\mathrm{op}}^2 \ge (RN^{-1}R^*)_{ii} \ge \frac{1}{\beta} (RQ^{-1}R^*)_{ii} = \frac{(K^{-1})_{ii} - 1}{\beta}.$$

The last identity is $RQ^{-1}R^* = K^{-1} - I$. In addition, $N \preceq \beta I$ implies $\langle e_i e_i^T, N^{-1} e_i e_i^T \rangle \geq 1/\beta$, giving the stated maximum. No assumption that ℓ_{ii} is bounded away from zero is needed.

Finally, for the stated decoder $D = Q^{-1}J^*T$, note $T^* = T^*\Pi$. Thus

$$\|D^* A_u\|_{\mathrm{F}}^2 \leq \frac{1}{\alpha}\|\Pi J Q^{-1} A_u\|_{\mathrm{F}}^2 = \frac{1}{\alpha}\langle A_u, Q^{-1} A_u \rangle,$$

so its noise amplification satisfies the same upper bound. This proves the decoder assertion as well as the equivalence. □

Every exact decoder has worst-case arbitrary-noise constant at least κ_{op}: data zero can represent H corrupted by $-HX$, while exactness forces the output for zero noiseless data to be zero. The ordinary least-squares inverse attains κ_{op} on the measurement range, extended by orthogonal projection. The local-whitening decoder is convenient but is not asserted to attain that exact optimum.

Capacity interpretation. For each vertex,

$$\frac{1}{(K^{-1})_{ii}} = \min_{a_i = 1} a^T K a.$$

This follows from Cauchy–Schwarz in the K inner product, with equality at $K^{-1}e_i/(K^{-1})_{ii}$. The relevant obstruction is a nonzero local amplitude that can be extended globally with very little compatibility energy. The criterion distinguishes this obstruction from both local frame singularity and a small global spectral gap whose eigenvectors are spread over many vertices.

Nonregular extension. For maximum degree at most q, assume full row rank for stencils of size at most q, and rank q for stencils of size $q+1$. Put $\ell_i = 0$ on the smaller stencils and use the Moore–Penrose local lift $(V_i^T)^\dagger$ there; the displayed inverse of $V_i^T V_i$ is not used when that matrix is singular. The same proof applies, with

$$a^T K a = \sum_{\deg i < q} a_i^2 + \tfrac{1}{2} \sum_{ij \in E} (\ell_{ij} a_i - \ell_{ji} a_j)^2.$$

The upper bound still uses $q+1$. This statement requires the stated local full-rank hypotheses; it is not a treatment of multiple null directions per row.

7 Two exact challenges to the generalized representation

7.1 A graph outside the original coloring and cycle assumptions

Use the Petersen graph on vertices $0, \ldots, 9$, with outer edges $i(i+1)$ modulo 5, spokes $i(i+5)$, and inner edges $(i+5)((i+2) \bmod 5 + 5)$. It is cubic and triangle-free. A locally injective four-coloring would require each color class to have $n/4$ vertices: each vertex's closed neighborhood has exactly one of each color, and each vertex belongs to four closed neighborhoods. Since $n = 10$, no such coloring exists.

Choose rows $X_i = (1, p_i^{(1)}, p_i^{(2)})/8$ with

$$(p_i)_{i=0}^9 = ((0,0),(1,2),(2,-1),(3,3),(-1,4),(-3,2),(4,-2),(-2,-3),(2,5),(-4,-1)).$$

All row norms are below one. Local null vectors are obtained from exact 3×3 cofactors. To avoid square roots in verification, let c_i be their primitive integer representatives and $d_i = \|c_i\|^2$. Form the rational/integer matrix

$$T_c = \sum_{ij \in E} (c_{ij} e_i - c_{ji} e_j)(c_{ij} e_i - c_{ji} e_j)^T.$$